

NIFTY 100 FINANCIAL INTELLIGENCE PLATFORM

Comprehensive Analyst & Technical Reference Guide (Sprint 6 Sign-Off)

Document Version:	1.0.0 (Production Release)
Target Audience:	Equity Research Analysts, Data Engineers, System Architects
Coverage Universe:	92 Nifty 100 Listed Companies across 10 Broad Sectors
Historical Period:	10 Financial Years (FY2014 – FY2024)
Core Frameworks:	Streamlit, FastAPI, SQLite, Scikit-Learn, ReportLab, Pandas
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1. Executive Overview & System Architecture

The Nifty 100 Financial Intelligence Platform is an enterprise-grade financial analytics suite engineered for equity research analysts, quantitative portfolio managers, and data engineers. It processes historical financial statements across 92 Nifty 100 companies covering 10 financial years (FY2014 to FY2024).

The architecture consists of a decoupled 5-tier pipeline: (1) Ingestion & ETL Normalization Engine, (2) SQLite Relational Store, (3) Analytics & ML Engine (Ratio engine, Screener, Peer percentiles, Valuation, KMeans clustering), (4) Presentation Layer (Streamlit Dashboard & PDF Generators), and (5) REST API Layer (FastAPI).

Key Design Rules: No data fabrication is allowed. Legitimate missing data (such as JIOFIN having only 2 years of history, ATGL missing cash flow records, or dividend yield being unavailable) is handled gracefully without artificial synthetic padding.

```
# System Architecture & Component Mapping
Data Ingestion (Excel/CSV) -> ETL Normalizer & DQ Rules -> SQLite (db/nifty100.db)
SQLite DB -> Analytics Modules (Ratio, Screener, Valuation, Peer, Clustering)
Analytics -> Streamlit GUI (8 Pages) + PDF Engine (Tearsheets) + FastAPI (16 Endpoints)
```

2. Data Model & SQLite Database Schema

The relational database db/nifty100.db contains 13 normalized tables. Enforced with Foreign Key constraints (PRAGMA foreign_keys = ON), the schema ensures structural integrity across companies, historical financial statements, documents, and derived analytical tables.

Primary Table Summary:

Table Name	Rows	Primary Keys	Description
companies	92	company_id	Master table of Nifty 100 companies with ticker & sector mapping
sectors	10	sector_id	10 broad sector classifications
profitandloss	1070	company_id, year	Historical income statement metrics (Sales, OPM, PAT, EPS)
balancesheet	1058	company_id, year	Historical balance sheet metrics (Assets, Equity, Borrowings)
cashflow	1056	company_id, year	Operating, Investing, and Financing Cash Flows
financial_ratios	1073	company_id, year	Derived KPIs (ROE, ROCE, D/E, FCF, CAGR, Quality Scores)
peer_percentiles	540	company_id, metric, year	0-1 percentile ranks across 11 peer groups

4. Financial Screener Engine & Custom Filters

The Financial Screener Engine evaluates universe companies against 15 financial metrics and assigns a sector-relative Composite Quality Score (0 to 100). Scores are computed using P10/P90 winsorization across broad sector groups.

Composite Weight Allocation: Profitability (35%), Cash Flow Quality (30%), Growth (20%), Leverage & Coverage (15%).

Special Rules Enforced: (1) Debt-to-Equity max filter is automatically waived for Banking & Financial sector firms. (2) Interest Coverage Ratio min filter treats Debt-Free companies as having infinite ICR. (3) Dividend yield filters strictly exclude NULL values.

```
# Screener Usage Example
from src.screener.engine import ScreenerEngine
engine = ScreenerEngine()
df = engine.load_universe_data(latest_year_only=True)
quality_compounders = engine.run_preset("quality_compounder", df=df)
```

5. Peer Percentile Rankings & Radar Analytics

The Peer Percentile Engine evaluates companies within 11 curated peer groups across 10 core financial metrics: ROE, ROCE, NPM, D/E (inverted), FCF, PAT CAGR 5yr, Revenue CAGR 5yr, EPS CAGR 5yr, Interest Coverage, and Asset Turnover.

Percentiles are calculated on a 0.0 to 1.0 scale using method='min' ranking. For leverage metrics like D/E, lower values receive higher percentiles.

Radar Chart Generation: Automated Matplotlib radar charts are stored in reports/radar_charts/ and integrated into company profile pages and tearsheet PDFs.

6. Valuation & Overvaluation Alert Engine

The Valuation Engine compares company valuation multiples (P/E, P/B, EV/EBITDA, FCF Yield) against historical 5-year medians and sector median benchmarks.

Overvaluation Alert Rules:

- CAUTION: Company P/E > 1.5x Sector Median P/E
- DISCOUNT: Company P/E < 0.7x Sector Median P/E
- FAIR: All other companies within normal valuation bands

Export Deliverables: output/valuation_summary.xlsx (all 92 companies) and output/valuation_flags.csv (caution & discount flagged companies).

7. KMeans Clustering & Machine Learning Archetypes

Using Scikit-Learn KMeans (n_clusters=5, random_state=42), all 92 companies are assigned to financial archetypes based on 5 standardized features: return_on_equity_pct, debt_to_equity, revenue_cagr_5yr, pat_cagr_5yr, and operating_profit_margin_pct.

Preprocessing: Missing values are imputed using sector medians followed by StandardScaler normalization.

Archetype Classifications:

1. High-Quality Compounders: Superior ROE (>20%), robust CAGR, low leverage.
2. Defensive Stalwarts: Moderate growth, stable margins, low debt.
3. Emerging Growth: High revenue expansion, reinvesting capital.
4. Value Cyclical: Capital intensive, moderate ROE.
5. Leveraged Turnaround: High D/E, recovering margins.

8. Cash Flow Intelligence & Capital Allocation Audit

The Cash Flow Engine evaluates earnings quality via CFO / PAT ratios, CapEx intensity (CapEx / Sales), and free cash flow conversion efficiency.

Quality Labels Assigned: High Quality (CFO/PAT ≥ 1.0), Moderate (0.7 - 1.0), Accrual Risk (< 0.7). Note: ATGL has zero cash flow records in source data and is legitimately labeled NaN without fabrication.

Capital Allocation Framework: Categorizes companies into Cash Accumulator, Growth Reinvestor, Debt Deleverager, or Shareholder Return focused based on 5-year aggregate cash flow deployment.

9. REST API Developer Documentation & OpenAPI Reference

FastAPI service exposes 16 endpoints across 8 routers under /api/v1 prefix with CORS and structured request logging.

Core API Routes:

- GET /api/v1/health — System health & DB table row counts
- GET /api/v1/companies — List 92 companies (sector/search filters)
- GET /api/v1/companies/{ticker} — Full profile + latest KPIs + cluster
- GET /api/v1/companies/{ticker}/tearsheet — Download 2-page PDF
- GET /api/v1/screener — Run screener with presets or custom filters
- GET /api/v1/sectors — 10 broad sectors with summary stats
- GET /api/v1/peers/{group_name} — Peer group percentiles
- GET /api/v1/portfolio/stats — P10 to P90 universe distributions

10. Quality Assurance, Pytest Suite & Acceptance Gates

The test suite contains 182 comprehensive unit, DQ, KPI, and API integration tests executed via Pytest. 0 failures.

Coverage: ETL Normalization, Loader functions, Profitability, Leverage, CAGR, DQ Rules, Screener, Peer Percentiles, and FastAPI endpoints.

HTML Test Report: Generated automatically at reports/pytest_report.html.

11. Deployment, CLI Utilities & Troubleshooting Guide

Running the Application:

- Launch Streamlit Dashboard: `streamlit run main.py`
- Launch FastAPI Server: `uvicorn src.api.main:app --port 8000`
- Execute Full Test Suite: `pytest --html=reports/pytest_report.html`
- Run ETL Pipeline: `python run_etl.py`
- Run Clustering & Stats: `python -m src.analytics.clustering`

Troubleshooting: If a company tearsheet returns 404, verify if it was legitimately skipped due to <3 years of history (e.g., JIOFIN).